

01:07

Prove using Resolution-Refutation Procedure Solved Example

- First, we convert the KB into clause form
- Converting KB1 to clause form:

$$\underline{(A \vee B) \wedge C} \Rightarrow E \Leftrightarrow \neg [(A \vee B) \wedge C] \vee E$$

$$a \rightarrow b \Rightarrow \neg a \vee b$$

- KB1: $(A \vee B) \wedge C \Rightarrow E$
- KB2: $D \wedge F \Rightarrow A$
- KB3: $\neg E \vee D$
- KB4: $B \wedge C$
- KB5: F

Prove A

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Prove using Resolution-Refutation Procedure Solved Example

- First, we convert the KB into clause form
- Converting KB1 to clause form:

$$\underline{(A \vee B) \wedge C} \Rightarrow E \Leftrightarrow \neg [(A \vee B) \wedge C] \vee E$$

$$\Leftrightarrow \neg(A \vee B) \vee \neg C \vee E$$

$$a \rightarrow b \Rightarrow \neg a \vee b$$

- KB1: $(A \vee B) \wedge C \Rightarrow E$
- KB2: $D \wedge F \Rightarrow A$
- KB3: $\neg E \vee D$
- KB4: $B \wedge C$
- KB5: F

Prove A

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Prove using Resolution-Refutation Procedure Solved Example

- First, we convert the KB into clause form
- Converting KB1 to clause form:

$$\begin{aligned} \underline{(A \vee B) \wedge C} \Rightarrow E &\Leftrightarrow \neg [(A \vee B) \wedge C] \vee E \\ &\Leftrightarrow \neg(A \vee B) \vee \neg C \vee E \\ &\Leftrightarrow (\neg A \wedge \neg B) \vee (\neg C \vee E) \end{aligned}$$

$$\begin{aligned} \checkmark \text{KB1.1: } \neg A \vee \neg C \vee E &\checkmark \\ \checkmark \text{KB1.2: } \neg B \vee \neg C \vee E &\checkmark \end{aligned}$$

$$a \rightarrow b \Rightarrow \neg a \vee b$$

$$\begin{aligned} \checkmark \text{KB1: } (A \vee B) \wedge C &\Rightarrow E \\ \text{KB2: } \underline{D \wedge F} &\Rightarrow A \\ \text{KB3: } \neg E \vee D & \\ \text{KB4: } B \wedge C & \\ \text{KB5: } F & \end{aligned}$$

Prove A

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Prove using Resolution-Refutation Procedure Solved Example

- First, we convert the KB into clause form
- Converting KB1 to clause form:

$$\begin{aligned} \underline{(A \vee B) \wedge C} \Rightarrow E &\Leftrightarrow \neg [(A \vee B) \wedge C] \vee E \\ &\Leftrightarrow \neg(A \vee B) \vee \neg C \vee E \\ &\Leftrightarrow (\neg A \wedge \neg B) \vee (\neg C \vee E) \end{aligned}$$

$$\begin{aligned} \checkmark \text{KB1.1: } \neg A \vee \neg C \vee E &\checkmark \\ \checkmark \text{KB1.2: } \neg B \vee \neg C \vee E &\checkmark \end{aligned}$$

$$a \rightarrow b \Rightarrow \neg a \vee b$$

$$\begin{aligned} \checkmark \text{KB1: } (A \vee B) \wedge C &\Rightarrow E \\ \text{KB2: } \underline{D \wedge F} &\Rightarrow A \\ \text{KB3: } \neg E \vee D & \\ \text{KB4: } B \wedge C & \\ \text{KB5: } F & \end{aligned}$$

Prove A

KB2: $\neg D \vee \neg F \vee A$

KB3: $\neg E \vee D$

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Prove using Resolution-Refutation Procedure Solved Example

- First, we convert the KB into clause form
- Converting KB1 to clause form:

$$\begin{aligned} \underline{(A \vee B) \wedge C} \Rightarrow E &\Leftrightarrow \neg [(A \vee B) \wedge C] \vee E \\ &\Leftrightarrow \neg(A \vee B) \vee \neg C \vee E \\ &\Leftrightarrow (\neg A \wedge \neg B) \vee (\neg C \vee E) \end{aligned}$$

✓ KB1.1: $\neg A \vee \neg C \vee E$ ✓

✓ KB1.2: $\neg B \vee \neg C \vee E$ ✓

KB2: $\neg D \vee \neg F \vee A$

KB3: $\neg E \vee D$

$a \rightarrow b \Rightarrow \neg a \vee b$

✓ KB1: $(A \vee B) \wedge C \Rightarrow E$

KB2: $D \wedge F \Rightarrow A$

KB3: $\neg E \vee D$

KB4: $B \wedge C$

KB5: F

Prove A

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Prove using Resolution-Refutation Procedure Solved Example

$\neg A$ resolve with KB2: $\neg D \vee \neg F \vee A$

KB1.1: $\neg A \vee \neg C \vee E$

KB1.2: $\neg B \vee \neg C \vee E$

KB2: $\neg D \vee \neg F \vee A$

KB3: $\neg E \vee D$

KB4.1: B

KB4.2: C

KB5: F

Prove A

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Prove using Resolution-Refutation Procedure Solved Example

$\neg A$ resolve with KB2: $\neg D \vee \neg F \vee A$
|
 $\neg D \vee \neg F$
|
 $\neg F \vee \neg E$ KB3: $\neg E \vee D$
KB5: F

KB1.1: $\neg A \vee \neg C \vee E$

KB1.2: $\neg B \vee \neg C \vee E$

KB2: $\neg D \vee \neg F \vee A$

KB3: $\neg E \vee D$

KB4.1: B

KB4.2: C

KB5: F

Prove A

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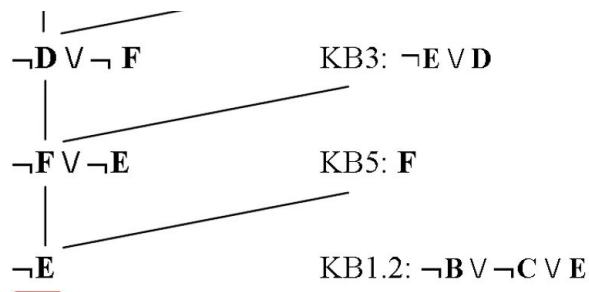
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Prove using Resolution-Refutation Procedure Solved Example

$\neg A$ resolve with KB2: $\neg D \vee \neg F \vee A$
|
 $\neg D \vee \neg F$

KB1.1: $\neg A \vee \neg C \vee E$

KB1.2: $\neg B \vee \neg C \vee E$



KB1.2: $\neg B \vee \neg C \vee E$
 KB2: $\neg D \vee \neg F \vee A$
 KB3: $\neg E \vee D$
 KB4.1: B
 KB4.2: C
 KB5: F

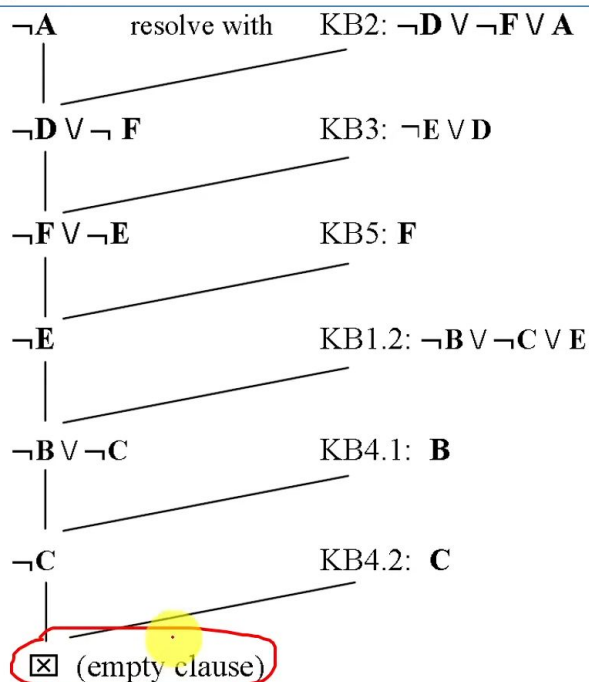
Prove A

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Prove using Resolution-Refutation Procedure Solved Example



KB1.1: $\neg A \vee \neg C \vee E$
 KB1.2: $\neg B \vee \neg C \vee E$
 KB2: $\neg D \vee \neg F \vee A$
 KB3: $\neg E \vee D$
 KB4.1: B
 KB4.2: C
 KB5: F

Prove A

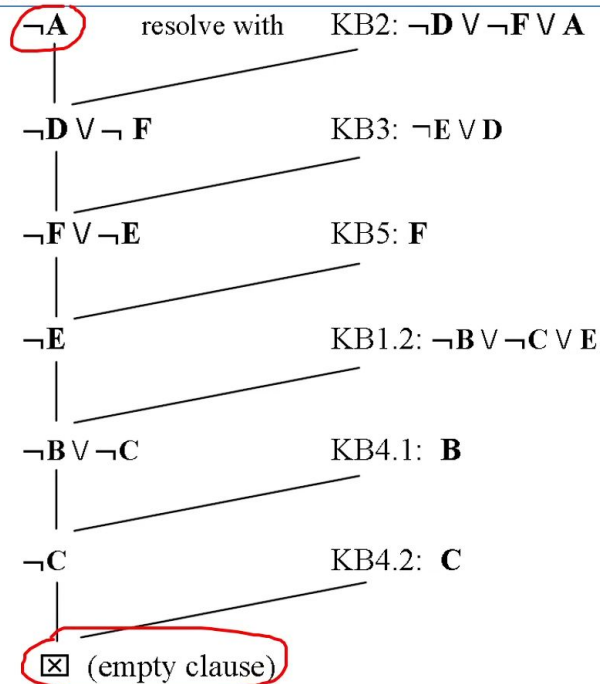
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Prove using Resolution-Refutation Procedure Solved Example

Prove using Resolution-Refutation Procedure Solved Example



KB1.1: $\neg A \vee \neg C \vee E$
 KB1.2: $\neg B \vee \neg C \vee E$
 KB2: $\neg D \vee \neg F \vee A$
 KB3: $\neg E \vee D$
 KB4.1: B
 KB4.2: C
 KB5: F

Prove A

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- Converting KB1 to clause form:

$$\begin{aligned}
 (A \vee B) \wedge C \Rightarrow E &\Leftrightarrow \neg [(A \vee B) \wedge C] \vee E \\
 &\Leftrightarrow \neg(A \vee B) \vee \neg C \vee E \\
 &\Leftrightarrow (\neg A \wedge \neg B) \vee (\neg C \vee E)
 \end{aligned}$$

KB1.1: $\neg A \vee \neg C \vee E$
 KB1.2: $\neg B \vee \neg C \vee E$
 KB2: $\neg D \vee \neg F \vee A$
 KB3: $\neg E \vee D$
 KB4.1: B
 KB4.2: C
 KB5: F

KB1: $(A \vee B) \wedge C \Rightarrow E$
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